Week 3 — Battery Maintenance, Chargers & Lithium

Sessions: 1 (Theory) · 2 (Lab) · **Total:** 4 hours
Primary LOs: LO-04, LO-05
Hands-on target: Session 2 ≈ 80% practical

Instructor Preparation Guide (Complete Day Before)

Pre-session setup

- [ ] Stage ferro-resonant and/or HF charger examples (onboard + portable if available)
- [ ] Prepare “no charge” training scenario (open AC, blown DC fuse, bad interlock, wrong pack voltage — pick 1–2)
- [ ] Lithium cart or bench pack ready with OEM app/tool if available; label chemistry (e.g., LiFePO4)
- [ ] Print `labs/W03_charger_troubleshoot.md` and `labs/W03_lithium_comparison.md`
- [ ] Confirm AC circuits for charger testing are GFCI-protected where required
- [ ] Post reminder: **never charge lithium with a lead-acid charger**

Planted faults

Scenario	Setup	Teachable point
No DC output	Open DC fuse or disconnected DC plug	Measure AC in, DC out systematically
Won’t finish charge	Sulfated bank or failed charge-complete sense (simulate)	Stages & sensing
Interlock open	Seat/charger interlock defeated training switch	Safety circuits matter
Wrong charger chemistry	Label-only demo (do not actually misuse)	Chemistry mismatch risk

Tools

DMM, clamp DC ammeter if available, IR thermometer, OEM lithium tool/app, charger manuals.

Week 3 Learning Objectives

1. Explain charger operating principles and stages (bulk, absorption, float/equalization).
2. Troubleshoot common charger failures: no output, low output, overcharge symptoms.
3. Describe key differences between lead-acid and lithium-ion golf cart systems.
4. Perform basic lithium diagnostics and explain BMS protective roles.

Session 1 — Theory & Demonstration (2 hours)

Materials needed: Ferro-resonant and/or HF charger samples; onboard vs portable examples; whiteboard for charge-stage curve; lithium pack or screenshot set; BMS app/tool if available; handout preview of charger isolation steps; PPE; DMM for demos; “never LA charger on lithium” poster note.

Timed Agenda

Time	Min	Activity	Instructor Notes
0:00	5	Safety: charging hydrogen + lithium thermal awareness	No smoking; ventilated charge area
0:05	15	Charger types: ferro-resonant taper vs HF multi-stage	Pros/cons for fleets
0:20	15	Charge stages & equalization purpose (flooded only)	**[DIAGRAM PLACEHOLDER: V/I vs time curve — bulk/absorption/float]**
0:35	10	Onboard vs portable; interlocks & inhibit circuits	Key/seat/charger plug logic varies by brand
0:45	5	Break	
0:50	20	Lithium advantages, risks, LiFePO4 vs generic “lithium” talk	Emphasize correct charger & BMS dependency
1:10	20	BMS functions: balance, OV/UV, temp, fault comms	Show sample SOC/cell screenshots if no live pack
1:30	15	Lead-acid maintenance review: watering, cleaning, torque	Tie to PM Week 10
1:45	10	Brand notes: Club Car / EZ-GO / Yamaha charger quirks overview	Always use model-year manual
1:55	5	Lab preview	

Key Teaching Points

- Charging algorithm must match chemistry — lead-acid EQ will damage lithium.
- “No charge” is a system problem: AC power, DC path, interlocks, pack acceptance, charger brain.
- BMS may disconnect for protection — treat as data, not a defective pack by default.

- DC-DC converters on lithium conversions power 12V accessories — separate from traction pack diagnostics.

Common Misconceptions

Misconception	Correction
“If the charger fan runs, it’s charging.”	Verify DC current into the pack.
“Lithium needs watering monthly.”	No watering; monitor SOC and use lithium charger only.
“Any 48V charger works on any 48V pack.”	Voltage class ≠ chemistry profile or connector/interlock logic.
“BMS faults mean replace the whole pack immediately.”	Read codes/history; check cell imbalance, temp, harness first.

Safety Reminders

- Disconnect power before opening charger cases (instructor-only internals unless designed for student access).
- Lithium thermal events: evacuate, extinguisher awareness, no water flood assumptions — follow shop EAP.
- Never bypass charger interlocks permanently.

Session 2 — Hands-On Lab (2 hours)

Materials needed: `labs/W03_charger_troubleshoot.md`, `labs/W03_lithium_comparison.md`; planted no-charge scenario; DMM (+ clamp if available); lithium demo cart/tool or data sheets; GFCI-protected AC as required.

Timed Agenda

Time	Min	Activity	Instructor Notes
0:00	5	Safety brief; assign charger vs lithium rotations	
0:05	25	**Station 1:** Measure AC input, DC output V, estimate/measure current	Complete charger worksheet Section A
0:30	25	**Station 2:** Diagnose planted “no charge / won’t finish”	Require written isolation path
0:55	5	Rotate / break	
1:00	25	**Station 3:** Lithium demo — cell voltages, SOC, fault history	If no lithium: structured video + data sheet exercise
1:25	20	Lead-acid vs lithium voltage profile comparison table	Teams fill comparison lab
1:45	15	Debrief + quiz preview; cleanup	

Assessment Focus

Charger troubleshooting worksheet + lithium vs lead-acid comparison quiz.

Week 3 Quiz (9 questions)

1. Bulk charge stage is best described as:
   A) Maintaining float only
   B) High current delivery until voltage setpoint approaches
   C) Equalizing tires
   D) Disabling the BMS

2. Equalization charges are appropriate for:
   A) All lithium packs weekly
   B) Flooded lead-acid when indicated by procedure
   C) AGM every night indefinitely
   D) Controllers

3. A common first measurement on a “no charge” complaint is:
   A) Transaxle oil
   B) AC power present at charger input
   C) Toe alignment
   D) Brush length

4. BMS cell balancing primarily helps:
   A) Keep cells within similar voltages for pack health
   B) Increase tire life

C) Bypass HPD forever
D) Replace the solenoid

5. True or False: Using a lead-acid charger on a lithium golf cart pack is acceptable if voltages match nominally.

Short Answer

1. Name three BMS protective functions.
2. List two differences between ferro-resonant and HF chargers.
3. Why do charger interlock circuits exist?
4. A lithium pack shows 0V at the main plug but the app shows SOC 55%. What is a likely explanation?

Answer Key

1. **B**
2. **B**
3. **B**
4. **A**
5. **False**
6. Any three: OV protection, UV protection, over-temp, under-temp charge inhibit, over-current, short protection, balancing, fault logging/comms
7. Examples: ferro = simpler taper/heavy; HF = multi-stage, lighter, smarter sensing / fault detection
8. Prevent drive-away while charging; ensure safe charge enable conditions
9. BMS contactor/MOSFET open due to protection or enable logic — pack not presenting voltage externally